

School of Natural Sciences and Mathematics

Molecular Biology and Healthcare Management (Double Major) (BS)

Bachelor of Science in Molecular Biology and Healthcare Management (Double Major)

[Degree Requirements](#) (152-153 semester credit hours)^{1, 2}

JSOM Faculty

FACG> jsom-healthcare-management-and-biology-bs

Professors: Ashiq Ali, Gary Bolton, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Kyle Hyndman, Varghese S. Jacob, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Stanley Liebowitz, Zhiang (John) Lin, Sumit K. Majumdar, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Mike W. Peng, Hasan Pirkul, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

Associate Professors: Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Livia Markoczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, Young U. Ryu, Harpreet Singh, Upender Subramanian, Shaojie Tang, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Yuan Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

Assistant Professors: Sheen Levine, Radha Mookerjee, Alejandro Rivera Mesias, Christian Von-Drathen, Guihua Wang, Steven Xiao

Associate Professors Emeriti: J. Richard Harrison, Jane Salk, David J. Springate

Visiting Professor: Emily Choi

Clinical Professors: John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Howard Dover, John Gamino, Randall S. Guttery, William Hefley, Marilyn Kaplan, Peter Lewin, Jeffrey Manzi, John F. McCracken, Diane S. McNulty, Daniel Rajaratnam, Kannan Ramanathan, Mark Thouin, Habte Woldu, Fang Wu, Laurie L. Ziegler

Clinical Associate Professors: Shawn Alborz, Dawn Owens, Carolyn Reichert

Clinical Assistant Professors: Moran Blueshtein, Jeffery (Jeff) Hicks, Kristen Lawson, Liping Ma, Ravi Narayan, Parneet Pahwa, Nassim Sohaee

Professors of Instruction: Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Matt Polze, Luell (Lou) Thompson

Associate Professors of Instruction: Judd Bradbury, Amal El-Ashmawi, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Jennifer G. Johnson, Hubert Zydorek

Assistant Professors of Instruction: Daniel Karnuta, Joseph Mauriello, Victoria D. McCrady

Professors of Practice: Tiffany A. Bortz, Alexander Edsel, Rajiv Shah, Keith Thurgood

Associate Professors of Practice: Richard Bowen, Jackie Kimzey, Margaret Smallwood, Steven Solcher, Kathy Zolton

Assistant Professors of Practice: Edward Meda, Timothy Stephens, Robert Wright

NSM Faculty

FACG> nsm-molecular-biology-and-healthcare-management-bs

Professors: Rockford K. Draper, Juan E. González, Kelli Palmer, Lawrence J. Reitzer, Stephen Spiro, Li Zhang, Michael Qiwei Zhang

Associate Professors: Joseph Boll, Mehmet Candas, Jeff L. DeJong, Nikki Delk, Tae Hoon Kim, Faruck Morcos, Duane D. Winkler, Zhenyu Xuan

Assistant Professors: Nicole De Nisco, Nicholas Dillon, Xintong Dong, Lin Jia, Purna Joshi, Jyoti Misra, Erica Sanchez, Darshan Sapkota

Professors Emeriti: Lee A. Bulla, Donald M. Gray

Associate Professors Emeriti: Gail A. M. Breen, Dennis L. Miller

Clinical Professor: David Murchison

Professors of Instruction: Scott A. Rippel, Uma Srikanth

Associate Professors of Instruction: Wen-Ju Lin, Elizabeth Pickett, Ilya Sapozhnikov

Assistant Professors of Instruction: Stephanie Boyd, Yi Huang, Meenakshi Maitra, Caitlin Maynard, Iti Mehta, Ramesh Padmanabhan, Jing Pan, Ruben D. Ramirez, Eva Sadat, Subha Sarcar, Michelle Wilson, Zhuoru Wu

Associate Professor of Practice: Uyen Henson

Research Assistant Professors: Li Liu, Ru-Hung Wang

Senior Lecturer: Wen-Ho Yu

UT Dallas Affiliated Faculty: Heng Du, Jung-whan (Jay) Kim

I. Core Curriculum Requirements: 42 semester credit hours³

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 2417](#) Calculus I^{4, 5, 6, 7, 8}

or [MATH 2413](#) Differential Calculus^{4, 5, 6, 7, 8}

Or select any 3 semester credit hours from [Mathematics Core](#) courses⁷ (see advisor)

Life and Physical Sciences: 6 semester credit hours

[CHEM 1311](#) General Chemistry I⁵

or [CHEM 1315](#) Honors Freshman Chemistry I⁵

[CHEM 1312](#) General Chemistry II⁵

or [CHEM 1316](#) Honors Freshman Chemistry II⁵

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Choose one of the following:⁹
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[BA 1310](#) Making Choices in Free Market Systems^{4, 5}
_ _

[BA 1320](#) Business in a Global World^{4, 5}
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[ECON 2301](#) Principles of Macroeconomics^{4, 5}
_ _

[ECON 2302](#) Principles of Microeconomics^{4, 5}
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Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two of the following:⁹
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[MATH 2419](#) Calculus II^{5, 6, 10}
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or [MATH 2414](#) Integral Calculus^{5, 6, 10}
_ _ _

[BA 1310](#) Making Choices in Free Market Systems^{4, 5}
_ _

[BA 1320](#) Business in a Global World^{4, 5}
_ _

[ECON 2301](#) Principles of Macroeconomics^{4, 5}
_ _

[ECON 2302](#) Principles of Microeconomics^{4, 5}
_ _

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 89-90 semester credit hours

Students must meet all course prerequisites.

Biology Major Preparatory Courses: 20-21 semester credit hours beyond Core Curriculum

[CHEM 1111](#) General Chemistry Laboratory I

or [CHEM 1115](#) Honors Freshman Chemistry Laboratory I

[CHEM 1112](#) General Chemistry Laboratory II

or [CHEM 1116](#) Honors Freshman Chemistry Laboratory II

[CHEM 1311](#) General Chemistry I⁵

or [CHEM 1315](#) Honors Freshman Chemistry I⁵

[CHEM 1312](#) General Chemistry II⁵

or [CHEM 1316](#) Honors Freshman Chemistry II⁵

[CHEM 2323](#) Introductory Organic Chemistry I⁴

or [CHEM 2327](#) Honors Organic Chemistry I⁴

[CHEM 2325](#) Introductory Organic Chemistry II⁴

or [CHEM 2328](#) Honors Organic Chemistry II⁴

[CHEM 2233](#) Introductory Organic Chemistry Laboratory⁴

or [CHEM 2237](#) Honors Organic Chemistry Laboratory⁴

[MATH 2417](#) Calculus I^{4, 5, 6, 7, 8}

or [MATH 2413](#) Differential Calculus^{4, 5, 6, 7, 8}

[MATH 2419](#) Calculus II^{5, 6, 10}

or [MATH 2414](#) Integral Calculus^{5, 6, 10}

[PHYS 2325](#) Mechanics and [PHYS 2125](#) Physics Laboratory I

or [PHYS 2421](#) Honors Physics I - Mechanics and Heat¹¹

[PHYS 2326](#) Electromagnetism and Waves

or [PHYS 2422](#) Honors Physics II - Electromagnetism and Waves

[PHYS 2126](#) Physics Laboratory II

Biology Core Courses: 33 semester credit hours

[BIOL 2111](#) Introduction to Modern Biology Workshop I⁴

[BIOL 2112](#) Introduction to Modern Biology Workshop II⁴

[BIOL 2281](#) Introductory Biology Laboratory⁴

[BIOL 2311](#) Introduction to Modern Biology I⁴

[BIOL 2312](#) Introduction to Modern Biology II⁴

[BIOL 3101](#) Classical and Molecular Genetics Workshop

[BIOL 3102](#) Eukaryotic Molecular and Cell Biology Workshop

[BIOL 3161](#) Biochemistry Workshop I

[BIOL 3162](#) Biochemistry Workshop II

[BIOL 3301](#) Classical and Molecular Genetics

[BIOL 3302](#) Eukaryotic Molecular and Cell Biology

[BIOL 3361](#) Biochemistry I

[BIOL 3362](#) Biochemistry II

or [BIOL 3335](#) Microbial Physiology

[BIOL 3380](#) Biochemistry Laboratory

[BIOL 4461](#) Biophysical Chemistry

Business Major Preparatory Courses: 12 semester credit hours beyond Core Curriculum

[ACCT 2301](#) Introductory Financial Accounting⁴

[ACCT 2302](#) Introductory Management Accounting⁴

[BLAW 2301](#) Business and Public Law⁴

[OPRE 3360](#) Managerial Methods in Decision Making Under Uncertainty

or [STAT 2332](#) Introductory Statistics for Life Sciences

or [STAT 3360](#) Probability and Statistics for Management and Economics

Choose two of the following⁹:

[BA 1310](#) Making Choices in Free Market Systems^{4, 5}

or [BA 1320](#) Business in a Global World^{4, 5}

or [ECON 2301](#) Principles of Macroeconomics^{4, 5}

or [ECON 2302](#) Principles of Microeconomics^{4, 5}

Business Core Courses: 24 semester credit hours

[BCOM 1300](#) Professional Communication¹²

or [BCOM 3300](#) Business Communication¹²

[BCOM 4300](#) Advanced Professional Communication

[FIN 3320](#) Business Finance

[IMS 3310](#) International Business

[ITSS 3300](#) Information Technology for Business

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

or [OBHR 3330](#) Introduction to Human Resource Management

[OPRE 3310](#) Supply Chain and Operations Management

III. Elective Requirements: 21 semester credit hours

A practicum experience of at least 160 working hours is required, with registration in one of the courses below.

[HMGH 4090](#) Healthcare Management Internship¹³

[BA 4090](#) Management Internship^{13, 14}

A community engagement experience is required, with registration in the course below.

[BA 4095](#) Social Sector Engagement and Community Outreach Practicum¹⁵

Healthcare Management Core Courses: 18 semester credit hours

[HMG 3301](#) Introduction to Healthcare Management

[HMG 3310](#) Healthcare Regulatory Environment

[HMG 3311](#) Healthcare Financial Analysis

[HMG 3320](#) Economics of Healthcare Environment

[HMG 4321](#) Introduction to Healthcare Information Systems

[HMG 4395](#) Capstone Senior Project - Healthcare Management

or [BPS 4395](#) Capstone Senior Project - Business

or [ENTP 4395](#) Capstone Senior Project - Entrepreneurship

Biology (3 semester credit hours):

[BIOL 4380](#) Cell and Molecular Biology Laboratory¹⁶

or [BIOL 3V96](#) Undergraduate Research in Molecular and Cell Biology

or [BIOL 4391](#) Senior Research in Molecular and Cell Biology

or [BIOL 4399](#) Senior Honors Research for Thesis in Molecular and Cell Biology

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Incoming freshmen must enroll and complete requirements of UNIV 1010 and the corresponding school-related freshman seminar course. Students, including transfer students, who complete their core curriculum at UT Dallas must take UNIV 2020.
2. Degree is 153-154 semester credit hours if students are required to take NATS 1101.
3. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
4. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
5. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
6. Six semester credit hours of Calculus are counted under Mathematics Core and Component Area Option Core, and 2 semester credit hours of Calculus are counted as Biology Major Preparatory Courses.

7. MATH 2413 is required for the Healthcare Management/Molecular Biology prescribed double major. Students may elect to substitute MATH 2417 for MATH 2413.
8. To make timely degree progress, students should complete MATH 2413 or MATH 2417 by the end of their first semester at UT Dallas. Students who will not meet this requirement should contact their academic advisor to discuss their degree timeline.
9. Certain courses listed are prerequisites for major core, major concentration, or major related courses. Choose accordingly.
10. Students may elect to substitute MATH 2419 for MATH 2414.
11. Students who complete PHYS 2421 do not need to complete PHYS 2125.
12. JSOM first-time-in-college freshmen are required to take BCOM 1300 in their first semester. Transfer students and students new to JSOM are required to take BCOM 3300 in their first semester.
13. Students may fulfill the internship requirement with HMGT 4090, BA 4090, or HMGT 4V90 (1-3 semester credit hours). The zero semester credit hour courses HMGT 4090 or BA 4090 are recommended as the most efficient way to satisfy this requirement.
14. Students employed in full-time jobs may fulfill their internship requirement with their current employer as a special project with preapproval from their program director
15. Students may fulfill the community engagement requirement with BPS 4396 or BA 4095, IMS 4335, ENTP 4340, or MKT 4360. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement.
16. Requires permission of the Biology Undergraduate Advisor to ensure training in recombinant DNA analysis.

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